

M5 Agentic AI - Market Research Team

1. Introduction

1.1. Lab Overview

In this lab, you will step into the role of a **technical AI lead at a fashion brand** preparing a summer sunglasses campaign. Your task is to design a **fully automated creative pipeline** that mirrors a real-world business scenario. Instead of handling each piece manually, you will guide a system that scans online sources for emerging fashion trends, matches those trends with sunglasses in the internal catalog, designs a campaign visual, generates a short marketing quote, and finally packages everything into an **executive-ready report**.

The goal is to experience how multiple agents, tools, and models can be orchestrated into a single, coherent workflow. By the end of this lab, you will have built a pipeline that feels less like a script of isolated steps and more like a small team working together to solve a creative challenge.

1.2. 🎯 Learning outcome

By completing this lab, you will see how to move beyond single-turn interactions with a model and instead design **multi-agent pipelines** that coordinate planning, research, and creative generation. You will learn how to ground agent reasoning in external tools so that outputs are not just imaginative but also supported by real data. You will also experiment with reflection and packaging steps that enforce quality control and prepare results for an executive audience.

In short, this lab is about learning how to combine the **imagination of large language models** with the **discipline of structured workflows**, giving you a practical pattern for building autonomous systems that are both creative and reliable.

2. Setup: Import libraries and load environment

As in previous labs, you now import the required libraries, load environment variables, and set up helper utilities.

```
In [1]: # =====  
# Imports  
# =====  
  
# --- Standard library ---
```

```

import base64
import json
import os
import re
from datetime import datetime
from io import BytesIO

# --- Third-party ---
import requests
import openai
from PIL import Image
from dotenv import load_dotenv
from IPython.display import Markdown, display
import aisuite

# --- Local / project ---
import tools
import utils

# =====
# Environment & Client
# =====
load_dotenv()
client = aisuite.Client()

```

3. Available Tools

Agentic pipelines only become effective when the model is given **explicit capabilities** beyond its base reasoning. Declaring these tools upfront makes the agent's action space unambiguous, ensures that prompts naturally guide tool selection, and keeps orchestration and testing transparent through well-defined interfaces.

You will assemble a **marketing research team**, a group of specialized agents collaborating to design a summer sunglasses campaign. To empower them, we start by defining the tools that will ground their reasoning in real data.

The first tool is `tools.tavily_search_tool`, which performs live web searches to uncover evidence of current fashion trends. Try it now by running a simple query for *"trends in sunglasses fashion"*:

```
In [2]: tools.tavily_search_tool('trends in sunglasses fashion')
```

```
Out[2]: [{'title': 'Sunglasses Trends 2026',
  'content': 'Sunglasses in 2026 reflect a perfect balance between statement aesthetics and enduring design. Bold frames, sculptural silhouettes, and retro influences lead the trends, with gradient lenses and metal shades taking center stage.\n\nSome styles of sunglasses are always in. Think sleek aviators, refined cat-eye frames, and classic square sunglasses. These iconic shapes transcend trends and continue to define modern elegance, season after season. [...] Timeless sunglasses combine minimal design with lasting impact. Think classic tortoiseshells, black frames, and metallic silhouettes. Whether you prefer a vintage or modern aesthetic, our timeless sunglasses collection offers styles that will never go out of style. [...] Butterfly Delicate styles with unexpected twists and intriguing subtlety. See the styles\n\nMaskA daring yet wearable choice for those who love a striking silhouette. See the styles\n\nOversizedThe grander the frame, the louder the fashion declaration. See the styles\n\n## Glasses Chains\n\nOnce purely practical, now unapologetically stylish. Glasses chains emerge as the year's most wanted statement. Golden chains and pearlescent details transform functionality into high-fashion whimsy.',
  'url': 'https://miaburton.com/en/sunglasses/trends'},
 {'title': 'Top Celebrity Sunglasses 2026: The Most-Worn Styles Right Now – Kraywoods',
  'content': 'One of the defining celebrity sunglasses trends of 2026 is the continued rise of slim, minimalist frames inspired by 90s and Y2K fashion. Narrow rectangular lenses and subtle oval metal frames dominate off-duty celebrity styling.\n\nThese micro sunglasses have been repeatedly seen on fashion-forward figures such as Bella Hadid, Hailey Bieber, and Kendall Jenner, reinforcing their influence within modern celebrity sunglasses culture. [...] Celebrity sunglasses continue to shape global eyewear trends year after year. From red carpet appearances to off-duty street style, sunglasses worn by celebrities consistently influence which frames dominate each season. In 2026, the strongest celebrity sunglasses trends reflect a balance between nostalgic minimalism, structured oversized silhouettes, tinted lenses, and architectural precision. [...] Cat-eye sunglasses remain a staple in celebrity sunglasses collections, consistently appearing at red carpet events and fashion weeks. In 2026, both bold oversized cat-eye frames and slimmer reinterpretations are trending.',
  'url': 'https://www.kraywoods.com/blogs/our-stories/top-celebrity-sunglasses'},
 {'title': '5 Sunglasses Trends That Will Make Your Whole Outfit This Summer',
  'content': 'New sunglasses trends, plus endless possibilities for your summer style persona. At brands like Celine, Khaite, Prada, and Loewe, shades arguably stole the spotlight. Ahead, discover the five sunglasses trends we're most excited to wear this summer. [...] Looking to make a style statement this summer? Start with sunglasses. Often regarded as an afterthought accessory—a piece that inspires less fanfare (and frenzy) than big-ticket bags or shoes—eyewear is actually a potent styling tool. The right sunglasses can level up your look. Extra-large shades might bring drama or mystery, while classic Wayfarers offer a timeless sense of cool. The spring 2026 runways delivered plenty of fresh inspiration, introducing a wave of exciting new sunglasses [...] Celine Spring 2026\n\nPascal Le Segretain/Getty Images Entertainment/Getty Images\n\nAviator-style acetate sunglasses\n\n$700\n\nBottega Veneta\n\nEclipse round-frame tortoiseshell acetate sunglasses\n\n$400\n\nMatteau\n\nRound-frame acetate sunglasses\n\n$525\n\nPrada\n\nGroupie\n\n$208\n\nBonnie Clyde\n\nChloé Wavy Sunglasses\n\n$525\n\nChloé\n\nThe Funk Daddy\n\n$99\n\nCrap Eyewear\n\n## The Athletic',
```

```
{
  'url': 'https://www.wmagazine.com/fashion/best-sunglasses-trends-2026'},
  {'title': 'The Best Sunglasses Trends for Men in 2026 | Vint & York',
   'content': "The metallic top brow bar trend is a variation of men's original flat brow style of sunglasses.\n\nThe top metal bar is a substitute for the large, straight acetate browline. This single change transformed the look into a more elegant and modern form. The sleek and refined design perfectly mirrors the current contemporary aesthetic.\n\nVint & York [...] 2026 men's glasses trends include both the new and the old. These vintage sunglasses talk the talk while you walk the walk. They are inspired by the steam-powered machines of the industrial revolution. Steampunk frames with metallic accents are a futuristic approach to a classic round frame.\n\nVint & York [...] Vint & York\n\nOne of the biggest trends you can get your hands on is the comeback of metal frames and clip on sunglasses. Out of many eyeglasses frames for men, they have made a considerable resurgence, quickly going from retro to modern hot. Men enjoy the slim lines and lightweight feel of metal frames, even slimmer and more lightweight than acetate frames.",
   'url': 'https://www.vintandork.com/blogs/content/best-men-sunglasses-trends'},
  {'title': 'The 5 Biggest Sunglasses Trends to Know for 2026 | Who What Wear',
   'content': "Designers presented a range of styles, such as dramatic, oversized silhouettes and wire frames, both of which recall sunglasses that were last widely popular in previous decades. The latest styles range from editorial and directional to practical and wearable, so opt for which version fits best into your personal wardrobe. Ahead, see the five biggest sunglasses trends of 2026, including the editor-approved styles to shop now.\n\n## Oversize Silhouettes [...] CELINE\n\nMetal Frame 28 Sunglasses in Metal\n\nMiu Miu, Oval Sunglasses\n\nMiu Miu\n\nOval Sunglasses\n\n## Acetate\n\nThick acetate frames are a running theme from the spring/summer 2026 collections. We saw everything from sleek black styles that feel classic yet futuristic to oversized silhouettes that have an exaggerated quality.\n\nSunglasses on the 2026 runway\n\n(Image credit: Launchmetrics)\n\nSunglasses on the 2026 runway\n\n(Image credit: Launchmetrics)\n\nKhaite x Oliver Peoples, Acetate Sunglasses [...] Khaite x Oliver Peoples\n\nAcetate Sunglasses\n\nLinda 53mm Butterfly Sunglasses\n\nTOM FORD\n\nLinda 53mm Butterfly Sunglasses\n\n## Modern Cat-Eyes\n\nCat-eye sunglasses are most commonly linked to '50s fashion, but we're now seeing them reimaged in new ways. At Ralph Lauren, they're finished with delicate wire frames, while at Bottega Veneta, they're sharply angled and outfitted with gradient lenses.\n\nSunglasses on the 2026 runway\n\n(Image credit: Launchmetrics)\n\nSunglasses on the 2026 runway",
   'url': 'https://www.whowhatwear.com/fashion/trends/sunglasses-trends-2026'}
}]
```

The second tool is `tools.product_catalog_tool`, which returns the internal sunglasses catalog. Each entry includes details such as product name, ID, description, stock quantity, and price. This structured data will allow the agents to connect online fashion trends with actual items in stock:

```
In [3]: tools.product_catalog_tool()
```

```
Out[3]: [{ 'name': 'Aviator',
  'item_id': 'SG001',
  'description': 'Originally designed for pilots, these teardrop-shaped lenses with thin metal frames offer timeless appeal. The large lenses provide excellent coverage while the lightweight construction ensures comfort during long wear.',
  'quantity_in_stock': 23,
  'price': 103},
  { 'name': 'Wayfarer',
  'item_id': 'SG002',
  'description': 'Featuring thick, angular frames that make a statement, these sunglasses combine retro charm with modern edge. The rectangular lenses and sturdy acetate construction create a confident look.',
  'quantity_in_stock': 6,
  'price': 92},
  { 'name': 'Mystique',
  'item_id': 'SG003',
  'description': 'Inspired by 1950s glamour, these frames sweep upward at the outer corners to create an elegant, feminine silhouette. The subtle curves and often embellished temples add sophistication to any outfit.',
  'quantity_in_stock': 3,
  'price': 88},
  { 'name': 'Sport',
  'item_id': 'SG004',
  'description': 'Designed for active lifestyles, these wraparound sunglasses feature a single curved lens that provides maximum coverage and wind protection. The lightweight, flexible frames include rubber grips.',
  'quantity_in_stock': 11,
  'price': 144},
  { 'name': 'Round',
  'item_id': 'SG005',
  'description': 'Circular lenses set in minimalist frames create a thoughtful, artistic appearance. These sunglasses evoke a scholarly or creative vibe while remaining effortlessly stylish.',
  'quantity_in_stock': 10,
  'price': 86}]
```

With these tools in place, you've defined a clear action space and reliable data sources. In the next section, you'll build the agents that use them to turn raw fashion signals into structured insights and campaign assets.

4. Agent Definitions — Building Your Team

Now that you have defined the tools, it's time to put them to work. In this phase, you will assemble a **marketing research team**, a group of specialized agents that you direct with natural instructions.

Each agent relies on the tools you introduced earlier, and together they transform raw trend data into a polished campaign report. We will define them one by one, introducing their role and showing the code that implements each.

4.1. Market Research Agent

With the **Market Research Agent**, you take the first step in building your campaign. You ask it to scan the web with `tavily_search_tool` and uncover what's trending in sunglasses fashion right now. Then you direct it to cross-check those signals against your internal catalog using `product_catalog_tool`, so you know which of your products fit the moment.

The agent hands back a concise brief: the top fashion insights it found, the products that align with them, and a short explanation of why those picks make sense for your summer push. This gives you a clear, data-driven foundation to shape the rest of the campaign.

You can now run the following cell to define the **Market Research Agent** in code.

```
In [4]: def market_research_agent(return_messages: bool = False):

    utils.log_agent_title_html("Market Research Agent", "🕵️")

    prompt_ = f"""
You are a fashion market research agent tasked with preparing a trend analysis.

Your goal:
1. Explore current fashion trends related to sunglasses using web search.
2. Review the internal product catalog to identify items that align with the trends.
3. Recommend one or more products from the catalog that best match emerging trends.
4. If needed, today date is {datetime.now().strftime("%Y-%m-%d")}.

You can call the following tools:
- tavily_search_tool: to discover external web trends.
- product_catalog_tool: to inspect the internal sunglasses catalog.

Once your analysis is complete, summarize:
- The top 2-3 trends you found.
- The product(s) from the catalog that fit these trends.
- A justification of why they are a good fit for the summer campaign.
"""

    messages = [{"role": "user", "content": prompt_}]
    tools_ = tools.get_available_tools()

    while True:
        response = client.chat.completions.create(
            model="openai:o4-mini",
            messages=messages,
            tools=tools_,
            tool_choice="auto"
        )

        msg = response.choices[0].message

        if msg.content:
            utils.log_final_summary_html(msg.content)
            return (msg.content, messages) if return_messages else msg.content
```

```

if msg.tool_calls:
    for tool_call in msg.tool_calls:
        utils.log_tool_call_html(tool_call.function.name, tool_call.
        result = tools.handle_tool_call(tool_call)
        utils.log_tool_result_html(result)

        messages.append(msg)
        messages.append(tools.create_tool_response_message(tool_call

else:
    utils.log_unexpected_html()
    return ("[⚠️ Unexpected: No tool_calls or content returned]", me

```

Let's try to get some advice from the **Market Research Agent** about our summer sunglasses campaign.

In [5]: `market_research_result = market_research_agent()`



Market Research Agent



Tool Call: tavily_search_tool

```

{"query": "2026 summer sunglasses fashion trends", "max_results": 5, "include_images": false}

```

✓ Tool Result:

[{'title': '2026', 'content': '| Categories | _General:_ Lists Timelines By year _Historical:_ 2020s By country Economics Politics Women\'s history Trends _Topical:_ Technology Transport Computing Science Health Arts and culture Art Paintings Sculptures Literature Music Fashion Sports Wrestling | Portal:Current events | [...] 164. ↑"Trump says interim accord with Iran to end war is "over"'. _Reuters_. July 8, 2026. Retrieved July 8, 2026.\n165. ↑"US says it has completed a new round of strikes on Iran". _Reuters_. July 8, 2026. Retrieved July 8, 2026.\n166. ↑"Singer\'s death brings Bangkok pub fire toll to 33". _Bangkok Post_. July 16, 2026. Archived from the original on July 17, 2026. Retrieved July 16, 2026. [...] 157. ↑Winfield, Nicole (June 30, 2026). "Pope begs breakaway traditionalist group to back off plan to consecrate its own bishops". _NBC News_. Associated Press. Retrieved June 30, 2026.\n158. ↑"Algerian Legislative Election, 2026". _PoliGlobe_. Archived from the original on March 24, 2026. Retrieved January 23, 2026.\n159. ↑"Iran begins week of funeral ceremonies for slain Supreme Leader Khamenei". _AlJazeera_. July 3, 2026. Retrieved July 3, 2026.\n160. ↑"Nations Championship", 'url': 'https://en.wikipedia.org/wiki/2026', {'title': 'August 4, 2026', 'content': '# Related Videos\n\n### Tuesday, August 4, 2026\n\n### gelmans bday\n\n### Shopping Day @ Live (8-4-2026)\n\n### Mason Gooding Drove Across the Country When He Moved to NYC\n\n### Mason Gooding Wore a Skirt for His "I Want Your Sex" Audition Tape\n\n### Jenna Dewan Talks About Her Stunt Mistake Filming the Season Finale of "The Rookie"\n\n### A Great Big World\'s Ian Axel – "Say Something" [...] Home\nEpisode Guide\nContests\nRecipes\nVideos\nStore\nTickets\nSHOW INFO\n+ Kelly Ripa\n+ Mark Consuelos\n+ Michael Gelman\n+ FAQ\nFollow LIVE\n\nclose\n\n### EPISODE GUIDE\n\nHome > Full Episode Guide\n\nAugust 4, 2026\n\nJENNA DEWAN of ABC\'s The Rookie drops by and MASON GOODING is here! Plus, "Summer Savings" and A GREAT BIG WORLD performs!\n\n### Share this:\n\nShare on Facebook (Opens in new window) Facebook\nShare on X (Opens in new window) X\nShare on Pinterest (Opens in new window) Pinterest', 'url': 'https://livelywithkellyandmark.com/guide/august-4-2026'}, {'title': '2026 in the United States', 'content': 'July 16\n+ 2026 Iran war: The White House maintains that Iran is still in talks with the US and hopes to negotiate an agreement. CENTCOM says it had expanded strikes into northern Iran on targets including military logistics infrastructure to "further degrade Iranian military capabilities".\n+ The Wisconsin Department of Health Services reports the first hospitalization of the summer in Wisconsin in connection with the cyclosporiasis outbreak.\nJuly 17 [...] 419. ↑ Somasekhar, Arathy (June 11, 2026). "Once an Arab oil embargo victim, US becomes world\'s top oil exporter". Reuters. Retrieved June 12, 2026.\n420. ↑ "Forbes Declares Elon Musk As The World\'s First Trillionaire". Forbes. June 12, 2026. Retrieved June 21, 2026.\n421. ↑ Chile, Patricio (June 12, 2026). "USMNT wins opening World Cup match against Paraguay 4-1". ABC News. Retrieved June 21, 2026. [...] 551. ↑ "Iran renews attacks on Gulf states after another night of US strikes". Reuters. July 17, 2026.\n552. ↑ Barrow, Bill; Stobbe, Mike (July 17, 2026). "Lettuce at Taco Bell in 5 states confirmed as a source of diarrhea-causing parasite". Associated Press. Retrieved July 22, 2026.\n553. ↑ "Investigation of 5-State Outbreak of Cyclospora Illnesses: Iceberg Lettuce (July 2026)". U.S. Food and Drug Administration. July 19, 2026. Retrieved July 22, 2026.', 'url': 'https://en.wikipedia.org/wiki/2026_in_the_United_States'}, {'title': 'Year 2026 Calendar – United States', 'content': 'a total solar eclipse, a near-total lunar eclipse, and the Northern Hemisphere\'s big summer meteor shower. [...] | Jan 1New

Year\'s Day Jan 19Martin Luther King Jr. Day Feb 14Valentine\'s Day Feb 16President\'s Day Mar 17St. Patrick\'s Day Apr 5Easter Sunday Apr 6Easter Monday.") Apr 15Tax Day so that their liability for income tax can be assessed.") May 5Cinco de MayoMay 10Mother\'s Day May 25 Memorial Day Jun 14Flag Day Jun 19Juneteenth Jun 21Father\'s Day Jul 3\'Independence Day\' day off Jul 4Independence Day Sep 7Labor Day Oct 12Columbus DayOct 31Halloween Nov 3Election Day Nov 11Veterans Day Nov [...] just hours after the Moon crosses the Sun in a total solar eclipse.

Image 70: Lone man hiking through bright red and green tundra with a glacier, glacier lake and snow-capped mountain in the background and a blue sky above. ##### The 11 Best Days to Take Off in Canada in 2026 Combine your days off with public holidays this year to create more free time for adventure.

Image 71: Sky Guide for August 2026 ##### Sky Guide for August 2026 August 2026 is a blockbuster month for skywatching: a total solar', 'url': 'https://www.timeanddate.com/calendar?country=1'}, {'title': '2026 Is Already Decided — What Will Happen', 'content': "fantastic. See you. [...] These are purely personal opinions and should not be taken as direct statements of fact about any individuals, whether stated explicitly or implied. Any opinion shared in this broadcast is drawn from referenced material specific to this publication only. Remember, what's discussed here are perspectives—not claims of fact. Copyright and all rights reserved. [...] the [music] fourth pressure point here. So moving on to number five. There is going to be an explosion of artificial content like video, audio, documents, screenshots, entire events that never really took place. [music] I would assume by the summer of 2026, the volume will be high enough that authenticity becomes really hard to prove in real life. [music] And that will absolutely change human behavior because once anything can be faked, everything can be denied. A compromising video comes up,", 'url': 'https://www.youtube.com/watch?v=XSN4uuL3jMg']}]

Tool Call: tavily_search_tool

```
{"query":"2026 summer sunglasses trends chunky frames tinted lenses mirrored street style","max_results":5}
```

✓ Tool Result:

[{'title': 'Sunglasses Trends for Summer 2026: Styles, Colors & Frames Everyone Will Wear', 'content': 'Lens color is a major focus in Summer 2026. Sunglasses with tinted lenses in yellow, amber, blue, and rose hues are everywhere. Gradient lenses are also trending, offering both style and functionality by reducing glare while maintaining visibility.\n\nTinted lenses add a relaxed, summery vibe and are especially popular for casual wear and outdoor dining.\n\nTrending lens colors:\n\nSoft yellow\nOcean blue\nSunset orange\nSmoky gray gradients\n\n# Transparent Frames and Eco-Inspired Materials [...] \$0.00 Cart\n\nMenu\n\nMenu\n\n# Reading Sunglasses & Polarized Bifocals for Men & Women\n\n### Reading With Sunglasses: Eyewear Trends & Sun Reader Tips\n\nHome » Uncategorized »\n\nUncategorized\n\n# Sunglasses Trends for Summer 2026: What Styles Everyone Will Be Wearing\n\nPosted by author-avatar Mass Vision Eyewear\n\nTrending sunglasses styles for Summer 2026 featuring oversized frames, tinted lenses, cat-eye designs, and wraparound sunglasses - Mass Vision\n\n10 Feb [...] Expect mirrored lenses, sharp contours, and lightweight frames designed for comfort and durability.\n\nPerfect for:\n\nOutdoor sports\nDriving and boating\nActive summer lifestyles\n\n## Retro Sunglasses with a Modern Twist\n\nClassic sunglasses shapes are being reimaged for Summer 2026. Aviators, round frames, and clubmaster-style designs are making a strong comeback, updated with thinner profiles, softer materials, and contemporary lens colors.', 'url': 'https://readingwithsunglasses.com/sunglasses-trends-summer-2026'}, {'title': 'Top Celebrity Sunglasses 2026: The Most-Worn Styles ...', 'content': 'Celebrity sunglasses continue to shape global eyewear trends year after year. From red carpet appearances to off-duty street style, sunglasses worn by celebrities consistently influence which frames dominate each season. In 2026, the strongest celebrity sunglasses trends reflect a balance between nostalgic minimalism, structured oversized silhouettes, tinted lenses, and architectural precision. [...] In 2026, the most popular celebrity sunglasses include slim 90s micro frames, oversized flat-top sunglasses, aviator sunglasses, cat-eye frames, shield silhouettes, geometric designs, and tinted lenses.\n\n## Are oversized sunglasses still in style?\n\nYes. Oversized sunglasses remain one of the strongest celebrity sunglasses trends, particularly structured square and flat-top designs frequently seen in celebrity street style.\n\n## Do tinted sunglasses provide UV protection? [...] Performers such as Dua Lipa and Bad Bunny have embraced lightly tinted lenses in both street style and stage appearances, highlighting the expressive side of celebrity sunglasses trends.\n\nUnlike darker lenses, tinted sunglasses soften contrast and add individuality without overpowering the face. When equipped with proper UV protection, they balance fashion with function — a key reason they continue rising among sunglasses worn by celebrities.', 'url': 'https://www.kraywoods.com/blogs/our-stories/top-celebrity-sunglasses'}, {'title': 'And Now, the Biggest Sunglasses Trends for 2026', 'content': 'CELINE\n\nMetal Frame 28 Sunglasses in Metal\n\nMiu Miu, Oval Sunglasses\n\nMiu Miu\n\nOval Sunglasses\n\n## Acetate\n\nThick acetate frames are a running theme from the spring/summer 2026 collections. We saw everything from sleek black styles that feel classic yet futuristic to oversized silhouettes that have an exaggerated quality.\n\nSunglasses on the 2026 runway\n\n(Image credit: Launchmetrics)\n\nSunglasses on the 2026 runway\n\n(Image credit: Launchmetrics)\n\nKhaite x Oliver Peoples, Acetate Sunglasses [...] Khaite x Oliver Peoples\n\nAcetate Sunglasses\n\nLinda 53mm Butterfly Sunglasses\n\nTOM FORD\n\nLinda 53mm Butterfly Sunglasses\n\n## Modern Cat-Eyes\n\nCat-eye sunglasses are most commonly linked to '50s f

fashion, but we're now seeing them reimagined in new ways. At Ralph Lauren, they're finished with delicate wire frames, while at Bottega Veneta, they're sharply angled and outfitted with gradient lenses.

Sunglasses on the 2026 runway (Image credit: Launchmetrics)

Sunglasses on the 2026 runway (Image credit: Launchmetrics)

Sunglasses on the 2026 runway (Image credit: Launchmetrics)

Miu Miu Runway Sunglasses

Miu Miu Runway Sunglasses

800 Sunglasses

800 Sunglasses

This post was previously published and has since been updated.

Explore More:

Kristen Nichols Associate Director, Special Projects, 'url': 'https://www.whowhatwear.com/fashion/trends/sunglasses-trends-2026', {'title': 'The Best Sunglasses Trends for Men in 2026', 'content': '2026 trends

The Best Sunglasses for Men in 2026: Popular Trending Styles

This summer's men's sunglasses styles and trends are a versatile bunch likely to turn quite a few heads. The cool shades on the market can upgrade any guy's look, giving him some serious style points.

Like most guys, you probably have one or two pairs of classic wayfarers lying around already. But summer is the perfect time to up your game with some fresh, bold frames!

Heavy browline sunglasses are another hot sunglasses trend for men. You'll find an even more prominent browline in these modern versions of the Clubmaster, giving them a thicker and manlier look.

The Vint & York Baker Sunglasses is a prime example of the thicker and manlier sunglasses frames. This retro-inspired, semi-rimless frame features square lenses and a heavy browline. The Baker men's sunglasses look like a square version of the Club master.

2026 men's glasses trends include both the new and the old. These vintage sunglasses talk the talk while you walk the walk. They are inspired by the steam-powered machines of the industrial revolution. Steampunk frames with metallic accents are a futuristic approach to a classic round frame.

The Vint & York, 'url': 'https://www.vintandnyork.com/blogs/content/best-men-sunglasses-trends', {'title': 'The Trending Sunglasses Styles for 2026 | Specsart.', 'content': 'Super-sized oversized sunglasses → Perfect for bold, carefree personalities who like their fashion loud and expressive.

Moderately oversized frames → Great for minimalists who want a little flair while keeping things polished and professional.

Rectangular Black Frame With Grey Tint Sunglasses-1

Rectangular Black Frame With Grey Tint Sunglasses-3

Black

EASTCOTE 1

£79.99

View

Tortoiseshell Sunglasses [...] Whether it's prescription glasses or sunglasses, tortoiseshell frames are staging a serious comeback in 2026. And honestly, if someone asks me to recommend a professional-looking eyewear style that isn't boring or too basic, tortoiseshell is always one of my go-to suggestions.

One sunglasses style that you couldn't miss on the runways this year is the futuristic shield frame. These visor-like sunglasses use a single sweeping lens across both eyes, connected only at the nose bridge. They completely hide your eyes and make you look like you're ready for any adventure (legal ones only, please avoid heists).', 'url': 'https://specsart.co.uk/blog/sunglasses-trends-2026'}]

 **Tool Call: product_catalog_tool**

```

{"max_items":10}
  
```

✔ Tool Result:

```
[{'name': 'Aviator', 'item_id': 'SG001', 'description': 'Originally designed for pilots, these teardrop-shaped lenses with thin metal frames offer timeless appeal. The large lenses provide excellent coverage while the lightweight construction ensures comfort during long wear.', 'quantity_in_stock': 23, 'price': 103}, {'name': 'Wayfarer', 'item_id': 'SG002', 'description': 'Featuring thick, angular frames that make a statement, these sunglasses combine retro charm with modern edge. The rectangular lenses and sturdy acetate construction create a confident look.', 'quantity_in_stock': 6, 'price': 92}, {'name': 'Mystique', 'item_id': 'SG003', 'description': 'Inspired by 1950s glamour, these frames sweep upward at the outer corners to create an elegant, feminine silhouette. The subtle curves and often embellished temples add sophistication to any outfit.', 'quantity_in_stock': 3, 'price': 88}, {'name': 'Sport', 'item_id': 'SG004', 'description': 'Designed for active lifestyles, these wraparound sunglasses feature a single curved lens that provides maximum coverage and wind protection. The lightweight, flexible frames include rubber grips.', 'quantity_in_stock': 11, 'price': 144}, {'name': 'Round', 'item_id': 'SG005', 'description': 'Circular lenses set in minimalist frames create a thoughtful, artistic appearance. These sunglasses evoke a scholarly or creative vibe while remaining effortlessly stylish.', 'quantity_in_stock': 10, 'price': 86}]
```

✓ Final Summary:

Here's the summer-2026 trend overview and my product picks:

1. Top Trends

a. Tinted & Gradient Lenses

– Soft pastels (yellow, rose, blue) and smoky-gray gradients are everywhere. They add a playful, laid-back summer vibe while still offering UV protection.

b. Oversized/Chunky Acetate Frames

– Bold silhouettes with thick, square or butterfly-style acetate are a key statement for the season.

c. Modern Retro Shapes (Cat-eyes & Shields)

– Reimagined '50s cat-eyes, aviator revivals and single-lens "shield"/wraparound styles blend nostalgia with a futuristic edge.

2. Catalog Matches

– Wayfarer (SG002)

• Thick acetate frame and angular, slightly oversized lenses echo the chunky-acetate statement trend. Ideal for street style and festival looks.

– Mystique (SG003)

• Upswept, cat-eye silhouette lines up perfectly with the modern-cat-eye revival. Its elegant curves suit both casual and dressy summer outfits.

– Sport (SG004)

• Wraparound one-piece lens mirrors the "shield"/visor trend. Lightweight, flexible and built for active, outdoor summer lifestyles (beach, boating, cycling).

3. Why These Work for Summer 2026

– All three can be fitted with tinted or gradient lenses to hit that color-lens trend.

– They cover the big directional shifts—statement acetate, retro-modern shapes and performance-driven shields—while remaining true to core catalog strengths (comfort, UV protection, durability).

– Together they give the campaign a versatile lineup that speaks to fashion-forward consumers and sporty, on-the-go users alike.

Next, you'll turn this brief into a visual concept with the Graphic Designer Agent.

4.2. Graphic Designer Agent

With the **Graphic Designer Agent**, you move from analysis to creativity.

You take the brief from your Market Research Agent and ask this one to translate it into a visual concept.

Because `aisuite` does not yet support direct image generation (like `gpt-image-1-mini`), you guide the process in two stages:

1. First, the agent uses `aisuite` with an OpenAI text model (`o4-mini`) to craft a vivid **prompt** and a short, engaging **caption**.
2. Then, the prompt is sent to OpenAI's `gpt-image-1-mini` API to generate the **campaign image** itself.

The result gives you everything you need to move forward: the generated image (saved locally for reuse), the exact prompt that produced it (useful for iteration), and a polished caption for campaign storytelling.

Note: At this point, `aisuite` does **not support direct image generation**. That's why you combine its text-based output (prompt + caption) with OpenAI's `gpt-image-1-mini` to produce the final campaign visual.

You can now run the following cell to define the **Graphic Designer Agent** in code.

```
In [6]: import base64

def graphic_designer_agent(trend_insights: str, caption_style: str = "short
    utils.log_agent_title_html("Graphic Designer Agent", "🎨")

    # Step 1: Generate prompt and caption using aisuite
    system_message = (
        "You are a visual marketing assistant. Based on the input trend insi
        "write a creative and visual prompt for an AI image generation model
    )
    user_prompt = f"""
Trend insights:
{trend_insights}
Please output:
1. A vivid, descriptive prompt to guide image generation.
2. A marketing caption in style: {caption_style}.
Respond in this format:
{{"prompt": "...", "caption": "..."}}
"""

    chat_response = client.chat.completions.create(
        model="openai:o4-mini",
        messages=[
            {"role": "system", "content": system_message},
            {"role": "user", "content": user_prompt}
        ]
    )
    content = chat_response.choices[0].message.content.strip()
    match = re.search(r'\{.*\}', content, re.DOTALL)
    parsed = json.loads(match.group(0)) if match else {"error": "No JSON ret
    prompt = parsed["prompt"]
    caption = parsed["caption"]

    # Step 2: Generate image with gpt-image-1-mini (returns base64, not URL)
    openai_client = openai.OpenAI()
    image_response = openai_client.images.generate(
        model="gpt-image-1-mini",
        prompt=prompt,
```

```

        size=size,
        quality="medium",    # gpt-image-1 accepts: low | medium | high | auto
        n=1,
        # no response_format - gpt-image-1 always returns b64_json
    )

    b64 = image_response.data[0].b64_json
    img_bytes = base64.b64decode(b64)
    img = Image.open(BytesIO(img_bytes))

    image_path = "generated_image.png"
    img.save(image_path)

    utils.log_final_summary_html(f"""
        <h3>Generated Image and Caption</h3>
        <p><strong>Image Path:</strong> <code>{image_path}</code></p>
        <p><strong>Generated Image:</strong></p>
        
        <p><strong>Prompt:</strong> {prompt}</p>
    """)

    return {
        "image_path": image_path,
        "prompt": prompt,
        "caption": caption,
    }

```

Now let's run the `graphic_designer_agent` to generate a campaign image, using the trend insights provided by the **Market Research Agent**.

```

In [7]: graphic_designer_agent_result = graphic_designer_agent(
        trend_insights=market_research_result,
    )

```



Graphic Designer Agent

✓ Final Summary:

<h3>Generated Image and Caption</h3>

<p>Image Path: <code>generated_image.png</code></p>

<p>Generated Image:</p>

<p>Prompt: An ultra-realistic editorial scene at golden-hour on a sun-kissed beach-boardwalk hybrid: three diverse models showcase the Summer 2026 eyewear lineup. Model one wears chunky, square acetate Wayfarer frames with a soft yellow-to-rose gradient lens. Model two strikes a pose in elegant Mystique cat-eye frames with smoky gray-to-blue fades. Model three powers through as a cyclist in a sleek Sport shield sunglasses tinted in pastel pink. Surround them with pastel beach umbrellas, swaying palm fronds, sun flares, and subtle motion blur of festival dancers and cyclists. Capture vibrant colors, crisp shadows, and a playful, laid-back summer vibe.</p>

With a visual in hand, you'll craft the campaign voice using the Copywriter Agent.

4.3. Copywriter Agent

Once the **Market Research Agent** and **Graphic Designer Agent** have done their work, you now turn to the **Copywriter Agent**. With both the campaign image and the trend summary in hand, you ask this agent to create the voice of your campaign.

It takes the visual and the analysis together as multimodal input and crafts a short, elegant marketing quote that captures the essence of the message. Along with the quote, it gives you a clear justification—why the phrase fits the image and how it ties back to the trends.

This way, you don't just get a catchy line, you also get the reasoning behind it, making it easier to defend and refine in front of stakeholders.

```
In [8]: def copywriter_agent(image_path: str, trend_summary: str, model: str = "openai:gpt-4o")
        """
        Uses aisuite (OpenAI only) to send an image and a trend summary and return a dict.

        Args:
            image_path (str): URL of the image to be analyzed.
            trend_summary (str): Text from the researcher agent.
            model (str): OpenAI model (e.g., openai:o4-mini, openai:gpt-4o)

        Returns:
            dict: {
                "quote": "...",
                "justification": "...",
            }
```



```

        "image_path": "...
    }
    """

    utils.log_agent_title_html("Copywriter Agent", "📝")

    # Step 1: Load local image and encode as base64
    with open(image_path, "rb") as f:
        img_bytes = f.read()

    b64_img = base64.b64encode(img_bytes).decode("utf-8")

    # Step 2: Build OpenAI-compliant multimodal message
    messages = [
        {
            "role": "system",
            "content": "You are a copywriter that creates elegant campaign c
        },
        {
            "role": "user",
            "content": [
                {
                    "type": "image_url",
                    "image_url": {
                        "url": f"data:image/png;base64,{b64_img}",
                        "detail": "auto"
                    }
                },
                {
                    "type": "text",
                    "text": f"""
Here is a visual marketing image and a trend analysis:

Trend summary:
\\\"\\\"\\\"{trend_summary}\\\"\\\"\\\"

Please return a JSON object like:
{{
    \"quote\": \"A short, elegant campaign phrase (max 12 words)\",
    \"justification\": \"Why this quote matches the image and trend\"
}}"""
                }
            ]
        }
    ]

    # Step 3: Send request via aisuite
    response = client.chat.completions.create(
        model=model,
        messages=messages,
    )

    # Step 4: Parse JSON response
    content = response.choices[0].message.content.strip()

    utils.log_final_summary_html(content)

```

```

try:
    match = re.search(r'\{.*\}', content, re.DOTALL)
    parsed = json.loads(match.group(0)) if match else {"error": "No valid JSON"}
except Exception as e:
    parsed = {"error": f"Failed to parse: {e}", "raw": content}

parsed["image_path"] = image_path
return parsed

```

Next, let's call the Copywriter Agent to generate a short campaign quote based on the marketing image and the trend insights produced earlier.

```

In [9]: copywriter_agent_result = copywriter_agent(
        image_path=graphic_designer_agent_result["image_path"],
        trend_summary=market_research_result,
    )

```



Copywriter Agent

✓ Final Summary:

```

{
  "quote": "Catch every summer ray with bold, tinted retro-chic shades.",
  "justification": "This line evokes the golden-hour beach scene and highlights the season's key trends—playful tinted/gradient lenses, statement acetate silhouettes and modern-retro shapes—just like the trio of Wayfarer, Mystique and Sport styles shown in the image."
}

```

With a quote and justification ready, you'll package everything into an executive-ready artifact using the Packaging Agent.

4.4. Packaging Agent

Finally, you bring in the **Packaging Agent** to tie everything together. After the **Market Research Agent**, **Graphic Designer Agent**, and **Copywriter Agent** have each contributed their part, this agent compiles the entire story into one polished artifact.

You ask it to take the trend summary, the campaign visual, the crafted quote, and the justification, and assemble them into an executive-ready markdown report. Along the way, it rewrites the trend insights for clarity and tone, ensures the quote is styled properly with the image, and organizes everything so the final document looks professional and persuasive.

With this step, you end up with a complete campaign package—easy to share, visually engaging, and ready for CEO-level review.

```
In [10]: def packaging_agent(trend_summary: str, image_url: str, quote: str, justification: str, output_path: str):
    """
    Packages the campaign assets into a beautifully formatted markdown report.

    Args:
        trend_summary (str): Summary of the market trends.
        image_url (str): URL of the campaign image.
        quote (str): Marketing quote to overlay.
        justification (str): Explanation for the quote.
        output_path (str): Path to save the markdown report.

    Returns:
        str: Path to the saved markdown file.
    """

    utils.log_agent_title_html("Packaging Agent", "📦")

    # We use this path in the src of the <img>
    styled_image_html = f"""
    ![Open the generated file to see]({image_url})
    """

    beautified_summary = client.chat.completions.create(
        model="openai:o4-mini",
        messages=[
            {"role": "system", "content": "You are a marketing communication expert. Please rewrite the following trend summary to be clear, professional, and engaging."},
            {"role": "user", "content": f"""
            \\\\"{trend_summary.strip()}\\\\"
            """
        ]
    ).choices[0].message.content.strip()

    utils.log_tool_result_html(beautified_summary)

    # Combine all parts into markdown
    markdown_content = f"""# 🕶 Summer Sunglasses Campaign – Executive Summary

## 📊 Refined Trend Insights
{beautified_summary}

## 🖼 Campaign Visual
{styled_image_html}

## 🗣 Campaign Quote
{quote.strip()}

## ✅ Why This Works
{justification.strip()}
    """
    return os.path.join(output_path, "campaign_package.md")
```

```

----
*Report generated on {datetime.now().strftime('%Y-%m-%d')}*
****

    with open(output_path, "w", encoding="utf-8") as f:
        f.write(markdown_content)

    return output_path

```

With your trend summary, campaign image, and quote ready, you now hand everything to the **Packaging Agent**. Its job is to pull these pieces together into a polished, executive-ready report. Run the next cell to generate it.

```

In [11]: packaging_agent_result = packaging_agent(
    trend_summary=market_research_result,
    image_url=graphic_designer_agent_result["image_path"],
    quote=copywriter_agent_result["quote"],
    justification=copywriter_agent_result["justification"],
    output_path=f"campaign_summary_{datetime.now().strftime('%Y-%m-%d_%H-%M-
)

```



Packaging Agent

✓ Tool Result:

Summer 2026 Trend & Product Summary

Overview

This summer's eyewear market centers on three defining trends:

1. Tinted & Gradient Lenses – Soft pastels (yellow, rose, blue) and smoky-gray fades deliver a playful, laid-back vibe while maintaining full UV protection.
2. Oversized/Chunky Acetate Frames – Bold, square and butterfly silhouettes in thick acetate make an unmistakable style statement.
3. Modern Retro Shapes – Reimagined '50s cat-eyes, aviator revivals and single-lens shield designs fuse nostalgic charm with futuristic appeal.

Product Selections

- Wayfarer (SG002) – A robust, angular acetate frame with slightly oversized lenses. Perfect for streetwear and festival goers seeking a bold, fashion-forward look.
- Mystique (SG003) – An elegant upswept cat-eye that transitions seamlessly from casual daytime to chic evening attire.
- Sport (SG004) – A lightweight, wraparound one-piece lens built for beach, boating and cycling, this model answers the call for performance-driven style.

Strategic Fit

- Lens Versatility: Each model accommodates the season's most sought-after tinted and gradient finishes.
- Trend Alignment: Together, they cover statement acetate, retro-modern silhouettes and active-lifestyle shields.
- Brand Strengths: All three reinforce our hallmarks—comfort, UV protection and durability—while appealing to both fashion-savvy consumers and on-the-move athletes.

This curated trio ensures a cohesive, versatile summer lineup that will resonate across style and activity segments, positioning us for a standout 2026 season.

The final result will be a beautifully formatted campaign report that you can view directly in the notebook. It will include:

- A refined trend summary that you will see rewritten for executive clarity
- A visually styled image with your campaign quote overlaid using HTML
- A clear justification so you understand why the visual and message align with current trends
- A timestamp showing you exactly when the report was generated

You can view it with:

```
In [12]: # Load and render the markdown content
with open(packaging_agent_result, "r", encoding="utf-8") as f:
```

```
md_content = f.read()
display(Markdown(md_content))
```

Summer Sunglasses Campaign – Executive Summary

Refined Trend Insights

Summer 2026 Trend & Product Summary

Overview

This summer's eyewear market centers on three defining trends:

1. Tinted & Gradient Lenses – Soft pastels (yellow, rose, blue) and smoky-gray fades deliver a playful, laid-back vibe while maintaining full UV protection.
2. Oversized/Chunky Acetate Frames – Bold, square and butterfly silhouettes in thick acetate make an unmistakable style statement.
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Product Selections

- Wayfarer (SG002) – A robust, angular acetate frame with slightly oversized lenses. Perfect for streetwear and festival goers seeking a bold, fashion-forward look.
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Strategic Fit

- Lens Versatility: Each model accommodates the season's most sought-after tinted and gradient finishes.
- Trend Alignment: Together, they cover statement acetate, retro-modern silhouettes and active-lifestyle shields.
- Brand Strengths: All three reinforce our hallmarks—comfort, UV protection and durability—while appealing to both fashion-savvy consumers and on-the-move athletes.

This curated trio ensures a cohesive, versatile summer lineup that will resonate across style and activity segments, positioning us for a standout 2026 season.

Campaign Visual



👉 Campaign Quote

Catch every summer ray with bold, tinted retro-chic shades.

✅ Why This Works

This line evokes the golden-hour beach scene and highlights the season's key trends—playful tinted/gradient lenses, statement acetate silhouettes and modern-retro shapes—just like the trio of Wayfarer, Mystique and Sport styles shown in the image.

Report generated on 2026-08-05

Finally, you'll wrap the entire workflow into a single callable function to run the full pipeline in one step.

5. Full Campaign Pipeline – run_sunglasses_campaign_pipeline

In this step, you will define a single function, `run_sunglasses_campaign_pipeline`, that ties all the pieces together into one seamless workflow for your summer sunglasses campaign.

The function will:

- Run market research to scan fashion trends and match them to your catalog.
- Generate a visually styled image and caption.
- Create a short, elegant campaign quote with justification.
- Package everything into a polished markdown report tailored for executive review.

By defining this function, you make it easy to run the **entire pipeline in one call** while still being able to trace intermediate results and view the final report.

```
In [13]: def run_sunglasses_campaign_pipeline(output_path: str = "campaign_summary.md")
        """
        Runs the full summer sunglasses campaign pipeline:
        1. Market research (search trends + match products)
        2. Generate visual + caption
        3. Generate quote based on image + trend
        4. Create executive markdown report

        Returns:
        dict: Dictionary containing all intermediate results + path to final report
        """
        # 1. Run market research agent
        trend_summary = market_research_agent()
        print("✅ Market research completed")

        # 2. Generate image + caption
        visual_result = graphic_designer_agent(trend_insights=trend_summary)
        image_path = visual_result["image_path"]
        print("🖼️ Image generated")

        # 3. Generate quote based on image + trends
        quote_result = copywriter_agent(image_path=image_path, trend_summary=trend_summary)
        quote = quote_result.get("quote", "")
        justification = quote_result.get("justification", "")
        print("💬 Quote created")

        # 4. Generate markdown report
        md_path = packaging_agent(
            trend_summary=trend_summary,
            image_url=image_path,
            quote=quote,
            justification=justification,
            output_path=f"campaign_summary_{datetime.now().strftime('%Y-%m-%d_%H-%M-%S')}.md")
```

```
)  
  
print(f"📦 Report generated: {md_path}")  
  
return {  
    "trend_summary": trend_summary,  
    "visual": visual_result,  
    "quote": quote_result,  
    "markdown_path": md_path  
}
```

You can now create a complete campaign report by running the pipeline in a single call.
Just execute the next cell:

```
In [14]: results = run_sunglasses_campaign_pipeline()
```



Market Research Agent



Tool Call: tavily_search_tool

```
{"query": "2026 summer sunglasses fashion  
trends", "max_results": 5, "include_images": false}
```

✓ Tool Result:

[{'title': '2026', 'content': '| Categories | _General:_ Lists Timelines By year _Historical:_ 2020s By country Economics Politics Women\'s history Trends _Topical:_ Technology Transport Computing Science Health Arts and culture Art Paintings Sculptures Literature Music Fashion Sports Wrestling | Portal:Current events | [...] 164. ↑ "Trump says interim accord with Iran to end war is "over"". _Reuters_. July 8, 2026. Retrieved July 8, 2026.\n165. ↑ "US says it has completed a new round of strikes on Iran". _Reuters_. July 8, 2026. Retrieved July 8, 2026.\n166. ↑ "Singer\'s death brings Bangkok pub fire toll to 33". _Bangkok Post_. July 16, 2026. Archived from the original on July 17, 2026. Retrieved July 16, 2026. [...] 157. ↑ Winfield, Nicole (June 30, 2026). "Pope begs breakaway traditionalist group to back off plan to consecrate its own bishops". _NBC News_. Associated Press. Retrieved June 30, 2026.\n158. ↑ "Algerian Legislative Election, 2026". _PoliGlobe_. Archived from the original on March 24, 2026. Retrieved January 23, 2026.\n159. ↑ "Iran begins week of funeral ceremonies for slain Supreme Leader Khamenei". _AlJazeera_. July 3, 2026. Retrieved July 3, 2026.\n160. ↑ "Nations Championship", 'url': 'https://en.wikipedia.org/wiki/2026', {'title': 'August 4, 2026', 'content': '# Related Videos\n\n### Tuesday, August 4, 2026\n\n### gelmans bday\n\n### Shopping Day @ Live (8-4-2026)\n\n### Mason Gooding Drove Across the Country When He Moved to NYC\n\n### Mason Gooding Wore a Skirt for His "I Want Your Sex" Audition Tape\n\n### Jenna Dewan Talks About Her Stunt Mistake Filming the Season Finale of "The Rookie"\n\n### A Great Big World\'s Ian Axel – "Say Something" [...] Home\nEpisode Guide\nContests\nRecipes\nVideos\nStore\nTickets\nSHOW INFO\n+ Kelly Ripa\n+ Mark Consuelos\n+ Michael Gelman\n+ FAQ\nFollow LIVE\n\nclose\n\n### EPISODE GUIDE\n\nHome > Full Episode Guide\n\nAugust 4, 2026\n\nJENNA DEWAN of ABC\'s The Rookie drops by and MASON GOODING is here! Plus, "Summer Savings" and A GREAT BIG WORLD performs!\n\n### Share this:\n\nShare on Facebook (Opens in new window) Facebook\nShare on X (Opens in new window) X\nShare on Pinterest (Opens in new window) Pinterest', 'url': 'https://livelywithkellyandmark.com/guide/august-4-2026'}, {'title': '2026 in the United States', 'content': 'July 16\n+ 2026 Iran war: The White House maintains that Iran is still in talks with the US and hopes to negotiate an agreement. CENTCOM says it had expanded strikes into northern Iran on targets including military logistics infrastructure to "further degrade Iranian military capabilities".\n+ The Wisconsin Department of Health Services reports the first hospitalization of the summer in Wisconsin in connection with the cyclosporiasis outbreak.\nJuly 17 [...] 419. ↑ Somasekhar, Arathy (June 11, 2026). "Once an Arab oil embargo victim, US becomes world\'s top oil exporter". Reuters. Retrieved June 12, 2026.\n420. ↑ "Forbes Declares Elon Musk As The World\'s First Trillionaire". Forbes. June 12, 2026. Retrieved June 21, 2026.\n421. ↑ Chile, Patricio (June 12, 2026). "USMNT wins opening World Cup match against Paraguay 4-1". ABC News. Retrieved June 21, 2026. [...] 551. ↑ "Iran renews attacks on Gulf states after another night of US strikes". Reuters. July 17, 2026.\n552. ↑ Barrow, Bill; Stobbe, Mike (July 17, 2026). "Lettuce at Taco Bell in 5 states confirmed as a source of diarrhea-causing parasite". Associated Press. Retrieved July 22, 2026.\n553. ↑ "Investigation of 5-State Outbreak of Cyclospora Illnesses: Iceberg Lettuce (July 2026)". U.S. Food and Drug Administration. July 19, 2026. Retrieved July 22, 2026.', 'url': 'https://en.wikipedia.org/wiki/2026_in_the_United_States'}, {'title': 'Year 2026 Calendar – United States', 'content': 'a total solar eclipse, a near-total lunar eclipse, and the Northern Hemisphere\'s big summer meteor shower. [...] | Jan 1New

Year\'s Day Jan 19Martin Luther King Jr. Day Feb 14Valentine\'s Day Feb 16President\'s Day Mar 17St. Patrick\'s Day Apr 5Easter Sunday Apr 6Easter Monday.") Apr 15Tax Day so that their liability for income tax can be assessed.") May 5Cinco de MayoMay 10Mother\'s Day May 25 Memorial Day Jun 14Flag Day Jun 19Juneteenth Jun 21Father\'s Day Jul 3\'Independence Day\' day off Jul 4Independence Day Sep 7Labor Day Oct 12Columbus DayOct 31Halloween Nov 3Election Day Nov 11Veterans Day Nov [...] just hours after the Moon crosses the Sun in a total solar eclipse.

Image 70: Lone man hiking through bright red and green tundra with a glacier, glacier lake and snow-capped mountain in the background and a blue sky above. ##### The 11 Best Days to Take Off in Canada in 2026 Combine your days off with public holidays this year to create more free time for adventure.

Image 71: Sky Guide for August 2026 ##### Sky Guide for August 2026 August 2026 is a blockbuster month for skywatching: a total solar', 'url': 'https://www.timeanddate.com/calendar?country=1'}, {'title': '2026 Is Already Decided — What Will Happen', 'content': "fantastic. See you. [...] These are purely personal opinions and should not be taken as direct statements of fact about any individuals, whether stated explicitly or implied. Any opinion shared in this broadcast is drawn from referenced material specific to this publication only. Remember, what's discussed here are perspectives—not claims of fact. Copyright and all rights reserved. [...] the [music] fourth pressure point here. So moving on to number five. There is going to be an explosion of artificial content like video, audio, documents, screenshots, entire events that never really took place. [music] I would assume by the summer of 2026, the volume will be high enough that authenticity becomes really hard to prove in real life. [music] And that will absolutely change human behavior because once anything can be faked, everything can be denied. A compromising video comes up,", 'url': 'https://www.youtube.com/watch?v=XSN4uuL3jMg'}]

Tool Call: tavily_search_tool

```
{"query":"2026 summer sunglasses trends chunky frames tinted lenses mirrored street style","max_results":5}
```


✓ Tool Result:

[{'title': 'Sunglasses Trends for Summer 2026: Styles, Colors & Frames Everyone Will Wear', 'content': 'Lens color is a major focus in Summer 2026. Sunglasses with tinted lenses in yellow, amber, blue, and rose hues are everywhere. Gradient lenses are also trending, offering both style and functionality by reducing glare while maintaining visibility.\n\nTinted lenses add a relaxed, summery vibe and are especially popular for casual wear and outdoor dining.\n\nTrending lens colors:\n\nSoft yellow\nOcean blue\nSunset orange\nSmoky gray gradients\n\n# Transparent Frames and Eco-Inspired Materials [...] \$0.00 Cart\n\nMenu\n\nMenu\n\n# Reading Sunglasses & Polarized Bifocals for Men & Women\n\n### Reading With Sunglasses: Eyewear Trends & Sun Reader Tips\n\nHome » Uncategorized »\n\nUncategorized\n\n# Sunglasses Trends for Summer 2026: What Styles Everyone Will Be Wearing\n\nPosted by author-avatar Mass Vision Eyewear\n\nTrending sunglasses styles for Summer 2026 featuring oversized frames, tinted lenses, cat-eye designs, and wraparound sunglasses - Mass Vision\n\n10 Feb [...] Expect mirrored lenses, sharp contours, and lightweight frames designed for comfort and durability.\n\nPerfect for:\n\nOutdoor sports\nDriving and boating\nActive summer lifestyles\n\n## Retro Sunglasses with a Modern Twist\n\nClassic sunglasses shapes are being reimaged for Summer 2026. Aviators, round frames, and clubmaster-style designs are making a strong comeback, updated with thinner profiles, softer materials, and contemporary lens colors.', 'url': 'https://readingwithsunglasses.com/sunglasses-trends-summer-2026'}, {'title': 'Top Celebrity Sunglasses 2026: The Most-Worn Styles ...', 'content': 'Celebrity sunglasses continue to shape global eyewear trends year after year. From red carpet appearances to off-duty street style, sunglasses worn by celebrities consistently influence which frames dominate each season. In 2026, the strongest celebrity sunglasses trends reflect a balance between nostalgic minimalism, structured oversized silhouettes, tinted lenses, and architectural precision. [...] In 2026, the most popular celebrity sunglasses include slim 90s micro frames, oversized flat-top sunglasses, aviator sunglasses, cat-eye frames, shield silhouettes, geometric designs, and tinted lenses.\n\n### Are oversized sunglasses still in style?\n\nYes. Oversized sunglasses remain one of the strongest celebrity sunglasses trends, particularly structured square and flat-top designs frequently seen in celebrity street style.\n\n### Do tinted sunglasses provide UV protection? [...] Performers such as Dua Lipa and Bad Bunny have embraced lightly tinted lenses in both street style and stage appearances, highlighting the expressive side of celebrity sunglasses trends.\n\nUnlike darker lenses, tinted sunglasses soften contrast and add individuality without overpowering the face. When equipped with proper UV protection, they balance fashion with function — a key reason they continue rising among sunglasses worn by celebrities.', 'url': 'https://www.kraywoods.com/blogs/our-stories/top-celebrity-sunglasses'}, {'title': 'And Now, the Biggest Sunglasses Trends for 2026', 'content': 'CELINE\n\nMetal Frame 28 Sunglasses in Metal\n\nMiu Miu, Oval Sunglasses\n\nMiu Miu\n\nOval Sunglasses\n\n## Acetate\n\nThick acetate frames are a running theme from the spring/summer 2026 collections. We saw everything from sleek black styles that feel classic yet futuristic to oversized silhouettes that have an exaggerated quality.\n\nSunglasses on the 2026 runway\n\n(Image credit: Launchmetrics)\n\nSunglasses on the 2026 runway\n\n(Image credit: Launchmetrics)\n\nKhaite x Oliver Peoples, Acetate Sunglasses [...] Khaite x Oliver Peoples\n\nAcetate Sunglasses\n\nLinda 53mm Butterfly Sunglasses\n\nTOM FORD\n\nLinda 53mm Butterfly Sunglasses\n\n## Modern Cat-Eyes\n\nCat-eye sunglasses are most commonly linked to '50s f

fashion, but we're now seeing them reimagined in new ways. At Ralph Lauren, they're finished with delicate wire frames, while at Bottega Veneta, they're sharply angled and outfitted with gradient lenses.

Sunglasses on the 2026 runway (Image credit: Launchmetrics)

Sunglasses on the 2026 runway (Image credit: Launchmetrics)

Miu Miu Runway Sunglasses

800 Sunglasses (Image credit: Launchmetrics)

This post was previously published and has since been updated.

Explore More:

Kristen Nichols, Associate Director, Special Projects, 'url': 'https://www.whowhatwear.com/fashion/trends/sunglasses-trends-2026', {'title': 'The Best Sunglasses Trends for Men in 2026', 'content': '2026 trends'}

The Best Sunglasses for Men in 2026: Popular Trending Styles

This summer's men's sunglasses styles and trends are a versatile bunch likely to turn quite a few heads. The cool shades on the market can upgrade any guy's look, giving him some serious style points.

Like most guys, you probably have one or two pairs of classic wayfarers lying around already. But summer is the perfect time to up your game with some fresh, bold frames!

Heavy browline sunglasses are another hot sunglasses trend for men. You'll find an even more prominent browline in these modern versions of the Clubmaster, giving them a thicker and manlier look.

The Vint & York Baker Sunglasses is a prime example of the thicker and manlier sunglasses frames. This retro-inspired, semi-rimless frame features square lenses and a heavy browline. The Baker men's sunglasses look like a square version of the Clubmaster.

2026 men's glasses trends include both the new and the old. These vintage sunglasses talk the talk while you walk the walk. They are inspired by the steam-powered machines of the industrial revolution. Steampunk frames with metallic accents are a futuristic approach to a classic round frame.

The Vint & York, 'url': 'https://www.vintandnyork.com/blogs/content/best-men-sunglasses-trends', {'title': 'The Trending Sunglasses Styles for 2026 | Specsart.', 'content': 'Super-sized oversized sunglasses → Perfect for bold, carefree personalities who like their fashion loud and expressive. Moderately oversized frames → Great for minimalists who want a little flair while keeping things polished and professional. Rectangular Black Frame With Grey Tint Sunglasses-1 Rectangular Black Frame With Grey Tint Sunglasses-3'}

Tortoiseshell Sunglasses [...] Whether it's prescription glasses or sunglasses, tortoiseshell frames are staging a serious comeback in 2026. And honestly, if someone asks me to recommend a professional-looking eyewear style that isn't boring or too basic, tortoiseshell is always one of my go-to suggestions.

One sunglasses style that you couldn't miss on the runways this year is the futuristic shield frame. These visor-like sunglasses use a single sweeping lens across both eyes, connected only at the nose bridge. They completely hide your eyes and make you look like you're ready for any adventure (legal ones only, please avoid heists).

'url': 'https://specsart.co.uk/blog/sunglasses-trends-2026'}

 **Tool Call: product_catalog_tool**

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{"max_items":10}
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✔ Tool Result:

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[{'name': 'Aviator', 'item_id': 'SG001', 'description': 'Originally designed for pilots, these teardrop-shaped lenses with thin metal frames offer timeless appeal. The large lenses provide excellent coverage while the lightweight construction ensures comfort during long wear.', 'quantity_in_stock': 23, 'price': 103}, {'name': 'Wayfarer', 'item_id': 'SG002', 'description': 'Featuring thick, angular frames that make a statement, these sunglasses combine retro charm with modern edge. The rectangular lenses and sturdy acetate construction create a confident look.', 'quantity_in_stock': 6, 'price': 92}, {'name': 'Mystique', 'item_id': 'SG003', 'description': 'Inspired by 1950s glamour, these frames sweep upward at the outer corners to create an elegant, feminine silhouette. The subtle curves and often embellished temples add sophistication to any outfit.', 'quantity_in_stock': 3, 'price': 88}, {'name': 'Sport', 'item_id': 'SG004', 'description': 'Designed for active lifestyles, these wraparound sunglasses feature a single curved lens that provides maximum coverage and wind protection. The lightweight, flexible frames include rubber grips.', 'quantity_in_stock': 11, 'price': 144}, {'name': 'Round', 'item_id': 'SG005', 'description': 'Circular lenses set in minimalist frames create a thoughtful, artistic appearance. These sunglasses evoke a scholarly or creative vibe while remaining effortlessly stylish.', 'quantity_in_stock': 10, 'price': 86}]
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✅ Final Summary:

Here's the summer-2026 trend overview and my product picks:

1. Top Trends

a. Tinted & Gradient Lenses

– Soft pastels (yellow, rose, blue) and smoky-gray gradients are everywhere. They add a playful, laid-back summer vibe while still offering UV protection.

b. Oversized/Chunky Acetate Frames

– Bold silhouettes with thick, square or butterfly-style acetate are a key statement for the season.

c. Modern Retro Shapes (Cat-eyes & Shields)

– Reimagined '50s cat-eyes, aviator revivals and single-lens "shield"/wraparound styles blend nostalgia with a futuristic edge.

2. Catalog Matches

– Wayfarer (SG002)

• Thick acetate frame and angular, slightly oversized lenses echo the chunky-acetate statement trend. Ideal for street style and festival looks.

– Mystique (SG003)

• Upswept, cat-eye silhouette lines up perfectly with the modern-cat-eye revival. Its elegant curves suit both casual and dressy summer outfits.

– Sport (SG004)

• Wraparound one-piece lens mirrors the "shield"/visor trend. Lightweight, flexible and built for active, outdoor summer lifestyles (beach, boating, cycling).

3. Why These Work for Summer 2026

– All three can be fitted with tinted or gradient lenses to hit that color-lens trend.

– They cover the big directional shifts—statement acetate, retro-modern shapes and performance-driven shields—while remaining true to core catalog strengths (comfort, UV protection, durability).

– Together they give the campaign a versatile lineup that speaks to fashion-forward consumers and sporty, on-the-go users alike.

✅ Market research completed



Graphic Designer Agent

✓ Final Summary:

<h3>Generated Image and Caption</h3>

<p>Image Path: <code>generated_image.png</code></p>

<p>Generated Image:</p>

<p>Prompt: An ultra-realistic editorial scene at golden-hour on a sun-kissed beach-boardwalk hybrid: three diverse models showcase the Summer 2026 eyewear lineup. Model one wears chunky, square acetate Wayfarer frames with a soft yellow-to-rose gradient lens. Model two strikes a pose in elegant Mystique cat-eye frames with smoky gray-to-blue fades. Model three powers through as a cyclist in a sleek Sport shield sunglasses tinted in pastel pink. Surround them with pastel beach umbrellas, swaying palm fronds, sun flares, and subtle motion blur of festival dancers and cyclists. Capture vibrant colors, crisp shadows, and a playful, laid-back summer vibe.</p>

 Image generated



Copywriter Agent

✓ Final Summary:

```
{
  "quote": "Catch every summer ray with bold, tinted retro-chic shades.",
  "justification": "This line evokes the golden-hour beach scene and highlights the season's key trends—playful tinted/gradient lenses, statement acetate silhouettes and modern-retro shapes—just like the trio of Wayfarer, Mystique and Sport styles shown in the image."
}
```

 Quote created



Packaging Agent

✓ Tool Result:

Summer 2026 Trend & Product Summary

Overview

This summer's eyewear market centers on three defining trends:

1. Tinted & Gradient Lenses – Soft pastels (yellow, rose, blue) and smoky-gray fades deliver a playful, laid-back vibe while maintaining full UV protection.
2. Oversized/Chunky Acetate Frames – Bold, square and butterfly silhouettes in thick acetate make an unmistakable style statement.
3. Modern Retro Shapes – Reimagined '50s cat-eyes, aviator revivals and single-lens shield designs fuse nostalgic charm with futuristic appeal.

Product Selections

- Wayfarer (SG002) – A robust, angular acetate frame with slightly oversized lenses. Perfect for streetwear and festival goers seeking a bold, fashion-forward look.
- Mystique (SG003) – An elegant upswept cat-eye that transitions seamlessly from casual daytime to chic evening attire.
- Sport (SG004) – A lightweight, wraparound one-piece lens built for beach, boating and cycling, this model answers the call for performance-driven style.

Strategic Fit

- Lens Versatility: Each model accommodates the season's most sought-after tinted and gradient finishes.
- Trend Alignment: Together, they cover statement acetate, retro-modern silhouettes and active-lifestyle shields.
- Brand Strengths: All three reinforce our hallmarks—comfort, UV protection and durability—while appealing to both fashion-savvy consumers and on-the-move athletes.

This curated trio ensures a cohesive, versatile summer lineup that will resonate across style and activity segments, positioning us for a standout 2026 season.

 Report generated: campaign_summary_2026-08-05_19-08-17.md

5.1. Results

Run the following cell to view the outputs generated by the full campaign pipeline.

```
In [15]: with open(results["markdown_path"], "r", encoding="utf-8") as f:
          md_content = f.read()
          display(Markdown(md_content))
```

Summer Sunglasses Campaign – Executive Summary

Refined Trend Insights

Summer 2026 Trend & Product Summary

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Campaign Visual



👉 Campaign Quote

Catch every summer ray with bold, tinted retro-chic shades.

✅ Why This Works

This line evokes the golden-hour beach scene and highlights the season's key trends—playful tinted/gradient lenses, statement acetate silhouettes and modern-retro shapes—just like the trio of Wayfarer, Mystique and Sport styles shown in the image.

Report generated on 2026-08-05

6. Key Takeaways

By completing this lab, you have seen how to:

- Use **multi-agent LLM pipelines** to automate a creative workflow end-to-end.
- Combine **reasoning, tool-calling, and external data** to ground your outputs in reality.
- Apply multimodal models (like `gpt-4o`) that process **both text and images** for tasks such as generating campaign quotes.
- Extend the model's abilities with tools (`tavily_search_tool` , `product_catalog_tool`) so your outputs are not only imaginative but also practical.
- Keep execution **transparent and debuggable** with structured logging and HTML-styled blocks.
- Deliver a polished, **executive-ready report** in Markdown format that blends insights, visuals, and justifications into a single artifact.

 **Congratulations!** 

Now you have successfully built and run a **multi-agent pipeline**: you researched trends, generated visuals, crafted a campaign quote, and packaged everything into an **executive-ready report**.

This workflow shows you how to combine the **creativity of LLMs** with the **discipline of structured orchestration**, giving you a repeatable pattern you can adapt to many real-world scenarios. 🌟

In []: